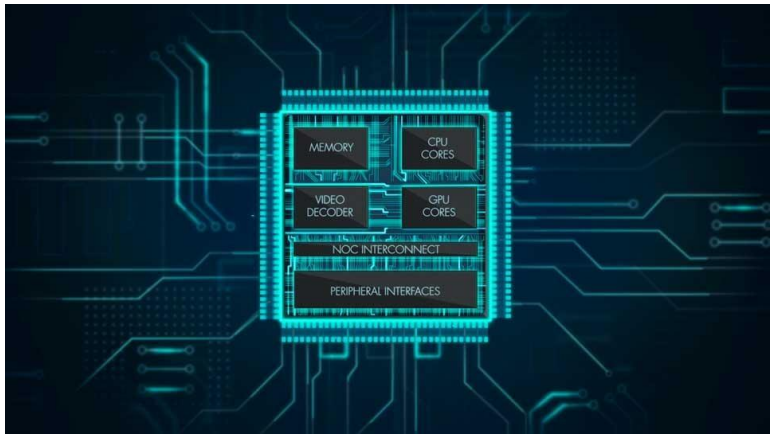
Digital System Design

Basic Practice 5



Date : 2026. 04. 23.

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Contents

- Dataflow
- Problem 1: Weight Stationary(WS) with Line Buffer
- Problem 2: Multi Channel Output Stationary(OS)
- Problem 3: Channel-wise(CW) Parallelism

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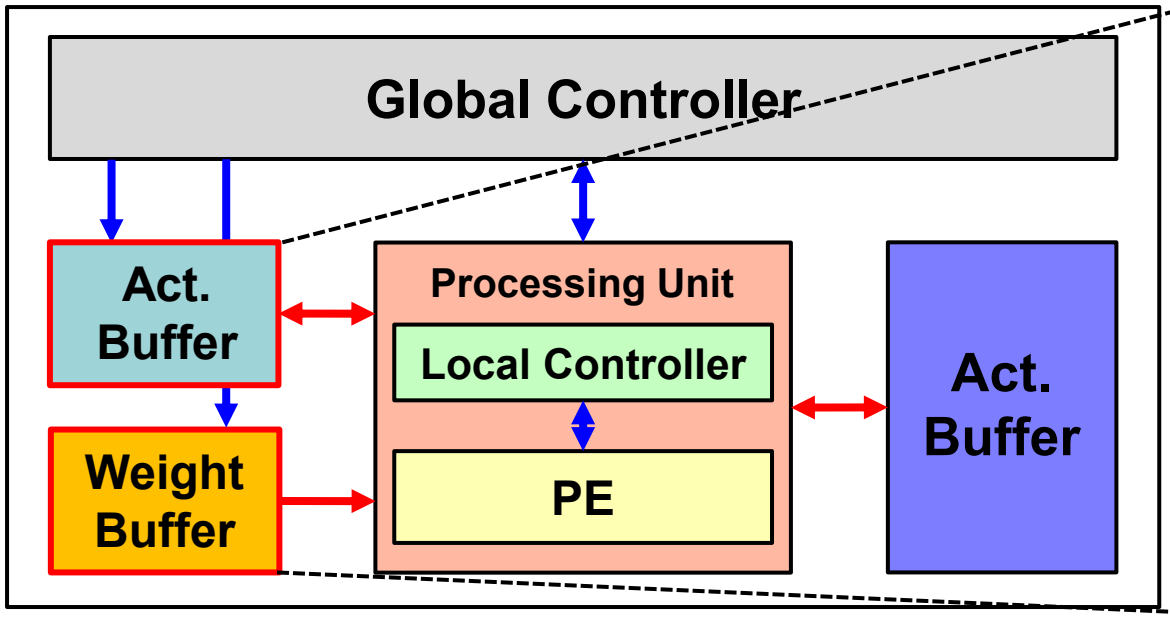
BRAM Data Fetch

BRAM Data Fetch

- Global controller determines how data is read from BRAM
- Data is fetched according to a predefined access pattern.



: Control : Data

Act. Buffer

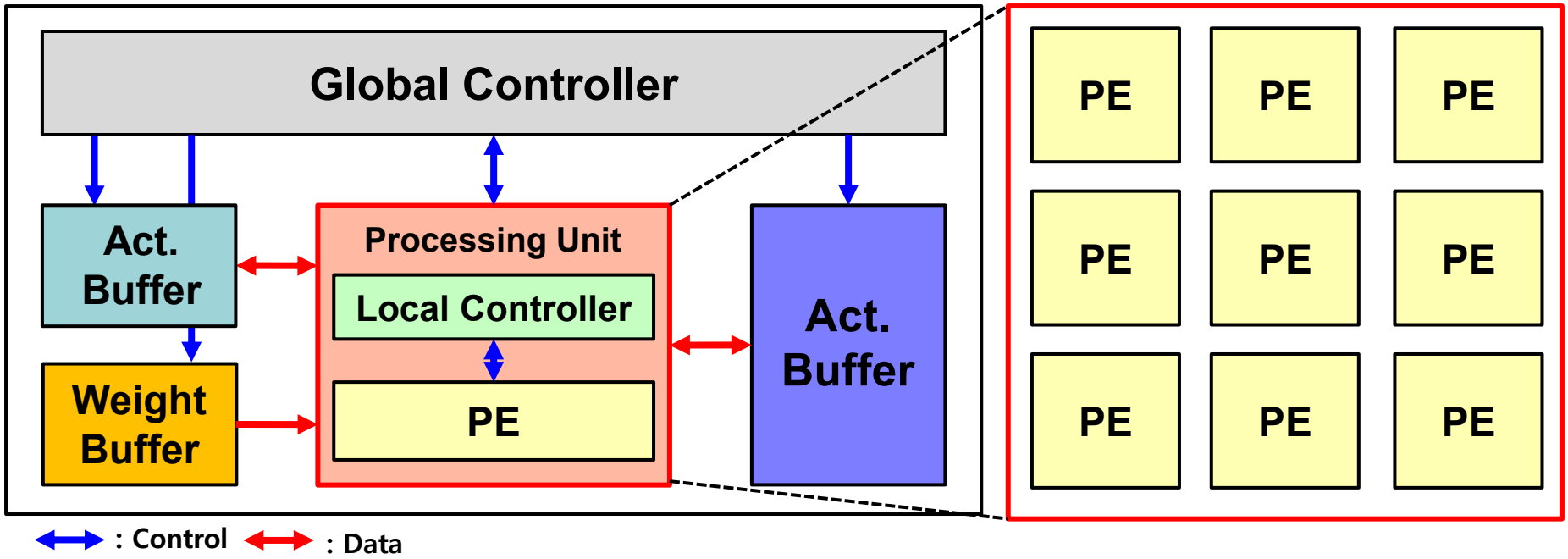
Addr.	Data
0	0x0001
1	0x0002
2	0x0003
3	0x0004
...	...
21	0x0016
22	0x0017
23	0x0018
24	0x0019

Weight Buffer

Addr.	Data
0	0x000B
1	0x000C
2	0x000D
3	0x000E
4	0x000F
5	0x0010
6	0x0011
7	0x0012
8	0x0013

Hardware Accelerator Design

- PU architecture is designed based on the dataflow.
- Computation structure follows the defined data movement.
- Optimal design depends on target performance and resource constraints.



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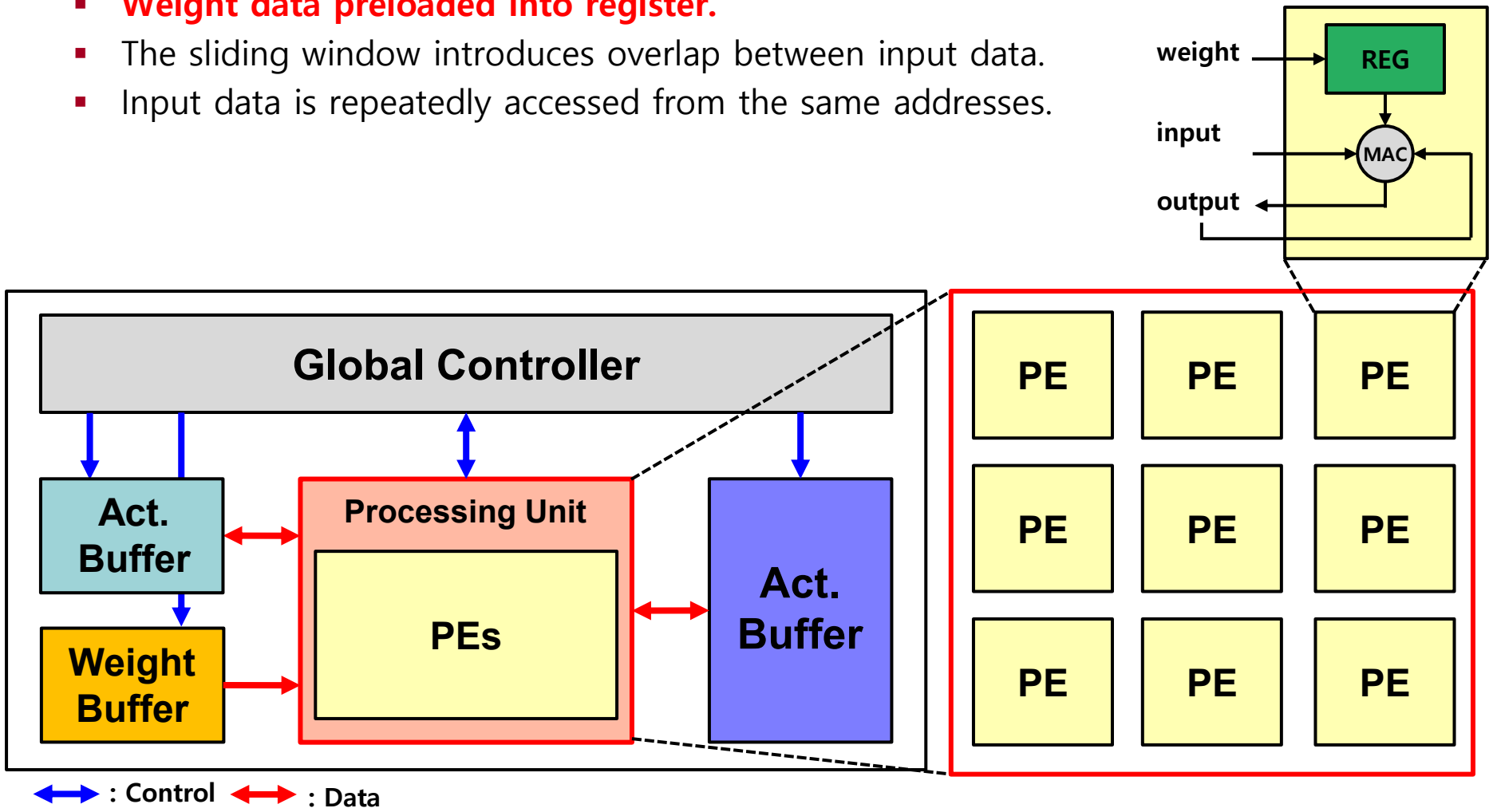
Prob 1. WS with Line Buffer (1)

Single Channel Weight Stationary

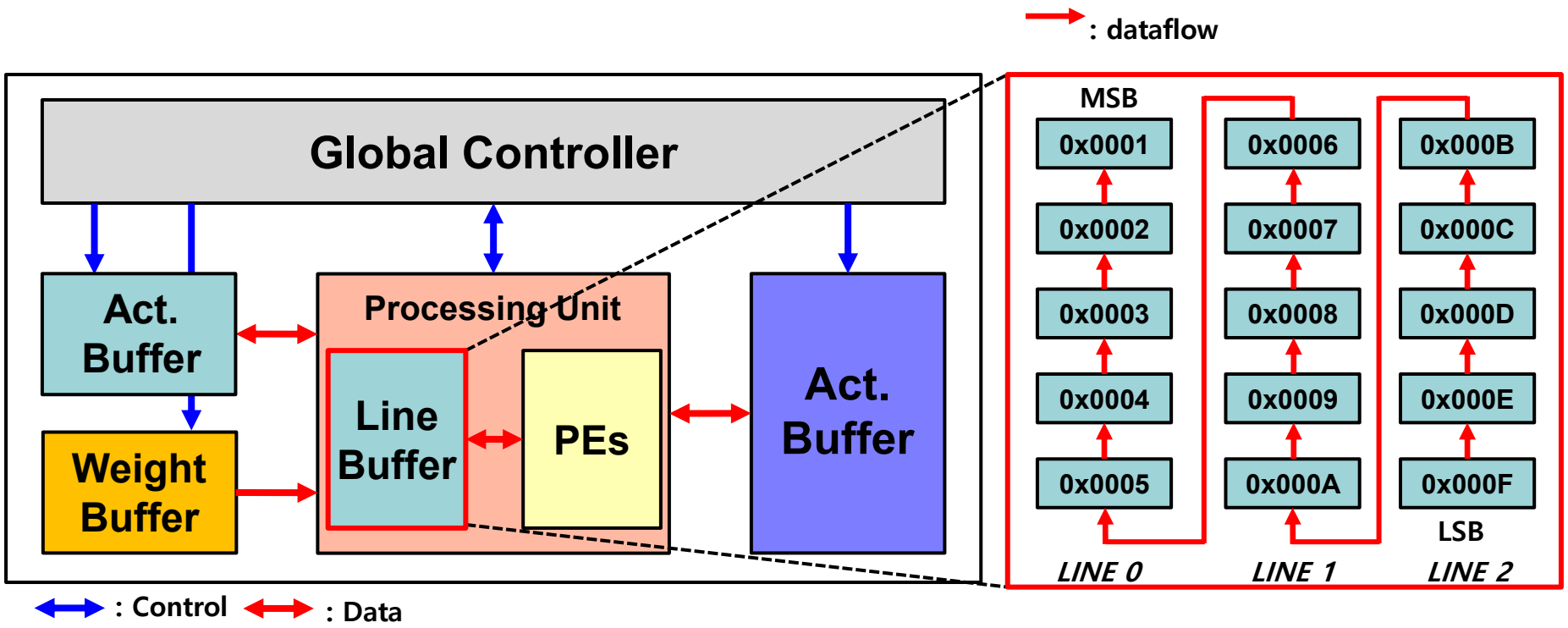
- **Weight data preloaded into register.**
- The sliding window introduces overlap between input data.
- Input data is repeatedly accessed from the same addresses.



Prob 1. WS with Line Buffer (2)

Single Channel Weight Stationary with Line Buffer

- **Line buffer stores input data using shift registers based on the valid signal.**
- Input data is loaded only once.

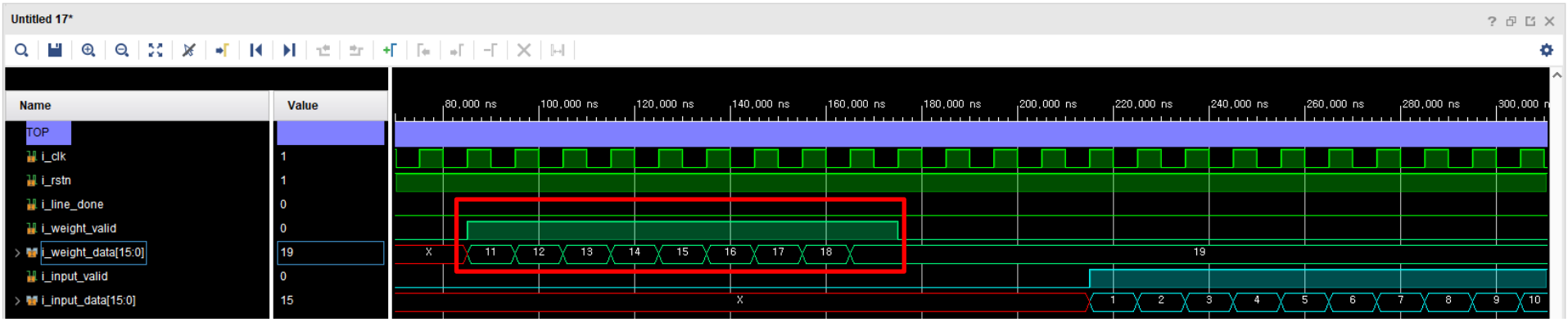


Prob 1. WS with Line Buffer (3)

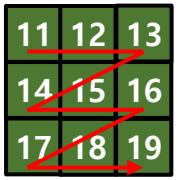
Waveform (1)

: dataflow

- Load weight data.



Load Weight

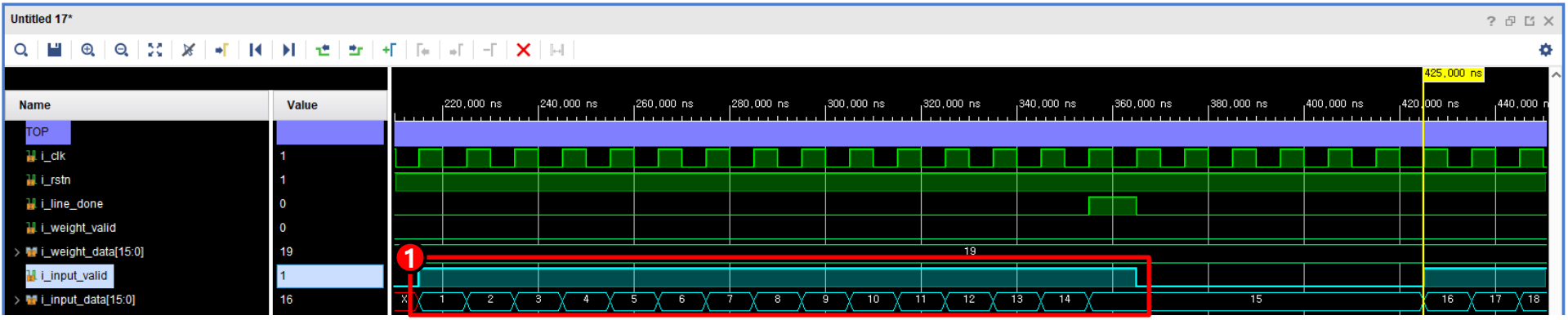


Prob 1. WS with Line Buffer (4)

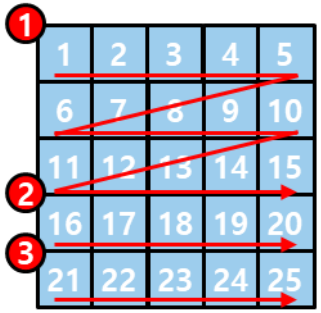
Waveform (2)

: dataflow

- Load input data.



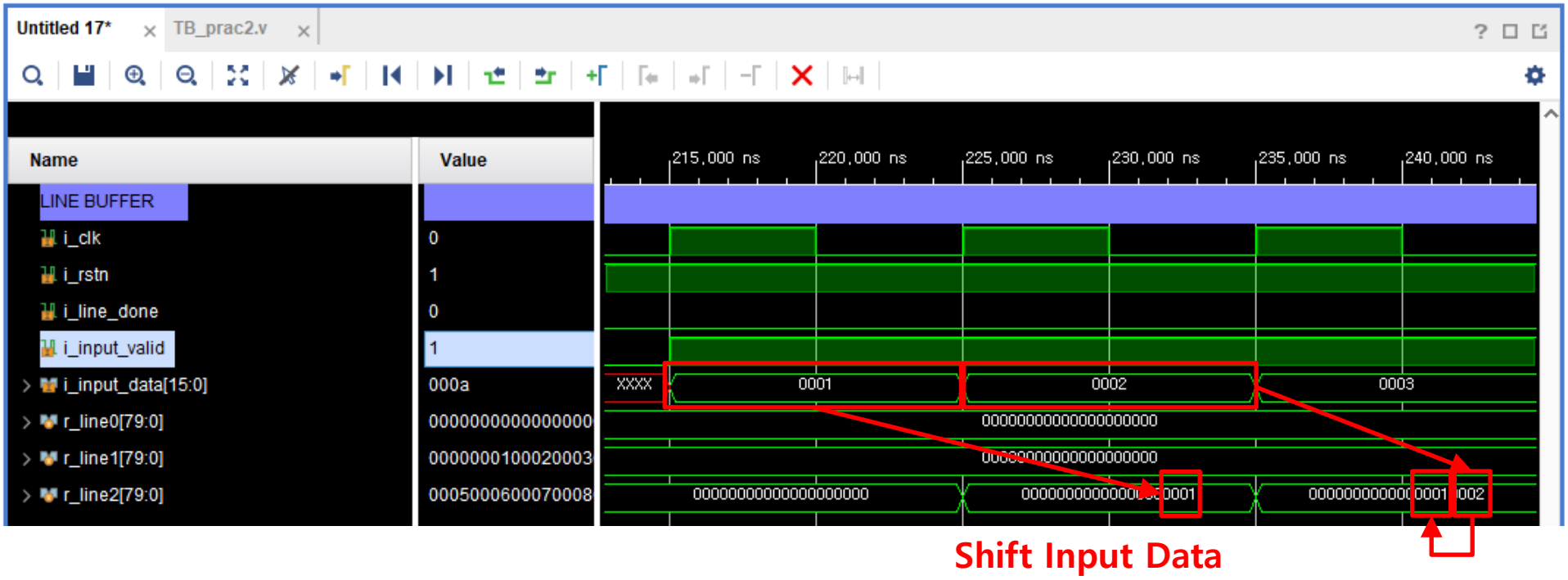
Load Input



Prob 1. WS with Line Buffer (5)

Waveform (3)

- Start saving input data by shifting line buffer.

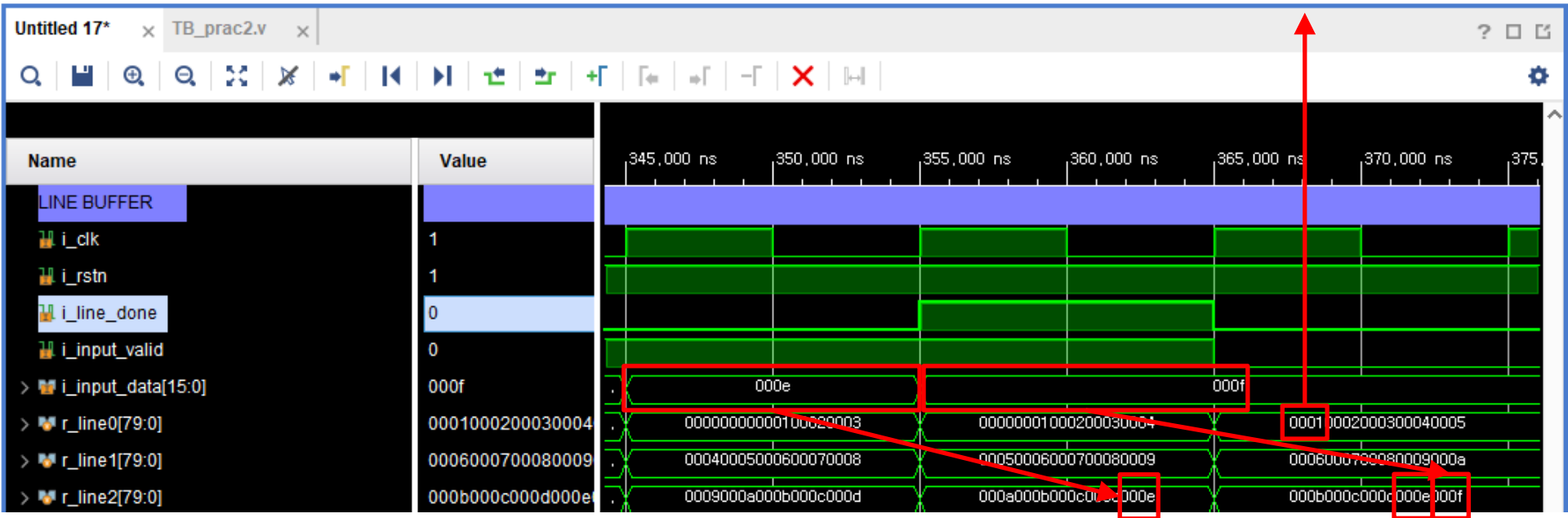


Prob 1. WS with Line Buffer (6)

Waveform (4)

- Save last data in line buffer.

First Input Data



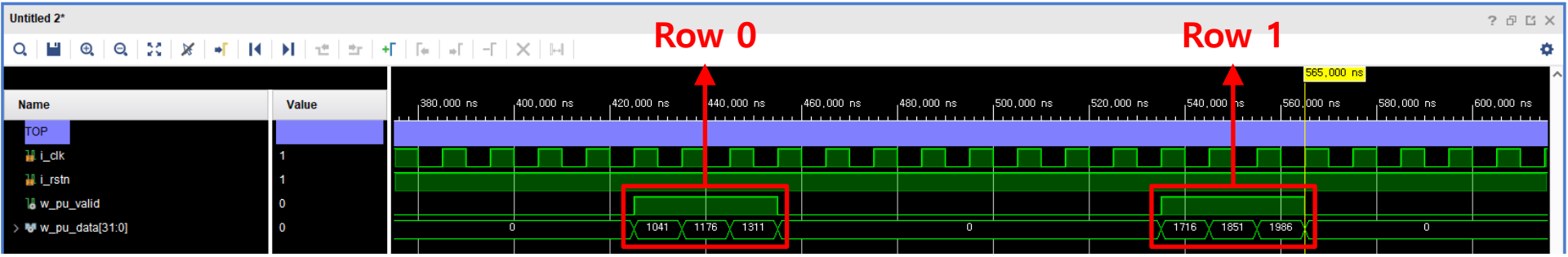
Shift Input Data

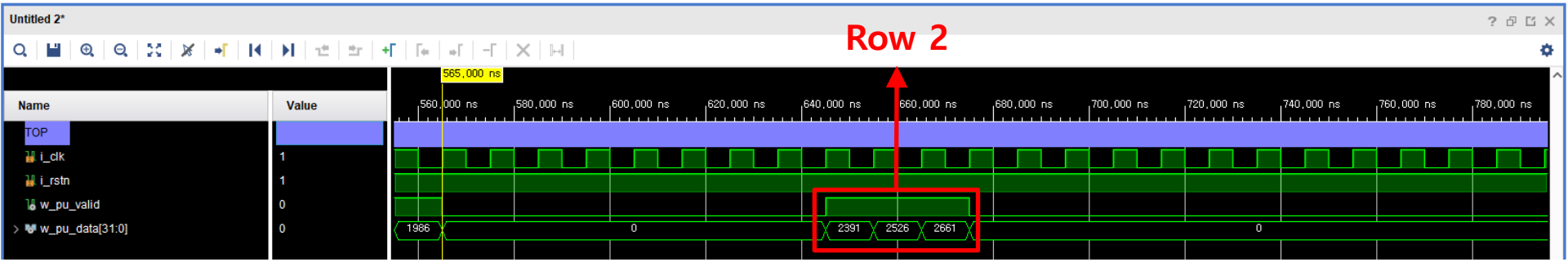
Prob 1. WS with Line Buffer (7)

Waveform (5)

- Output of Conv2d.

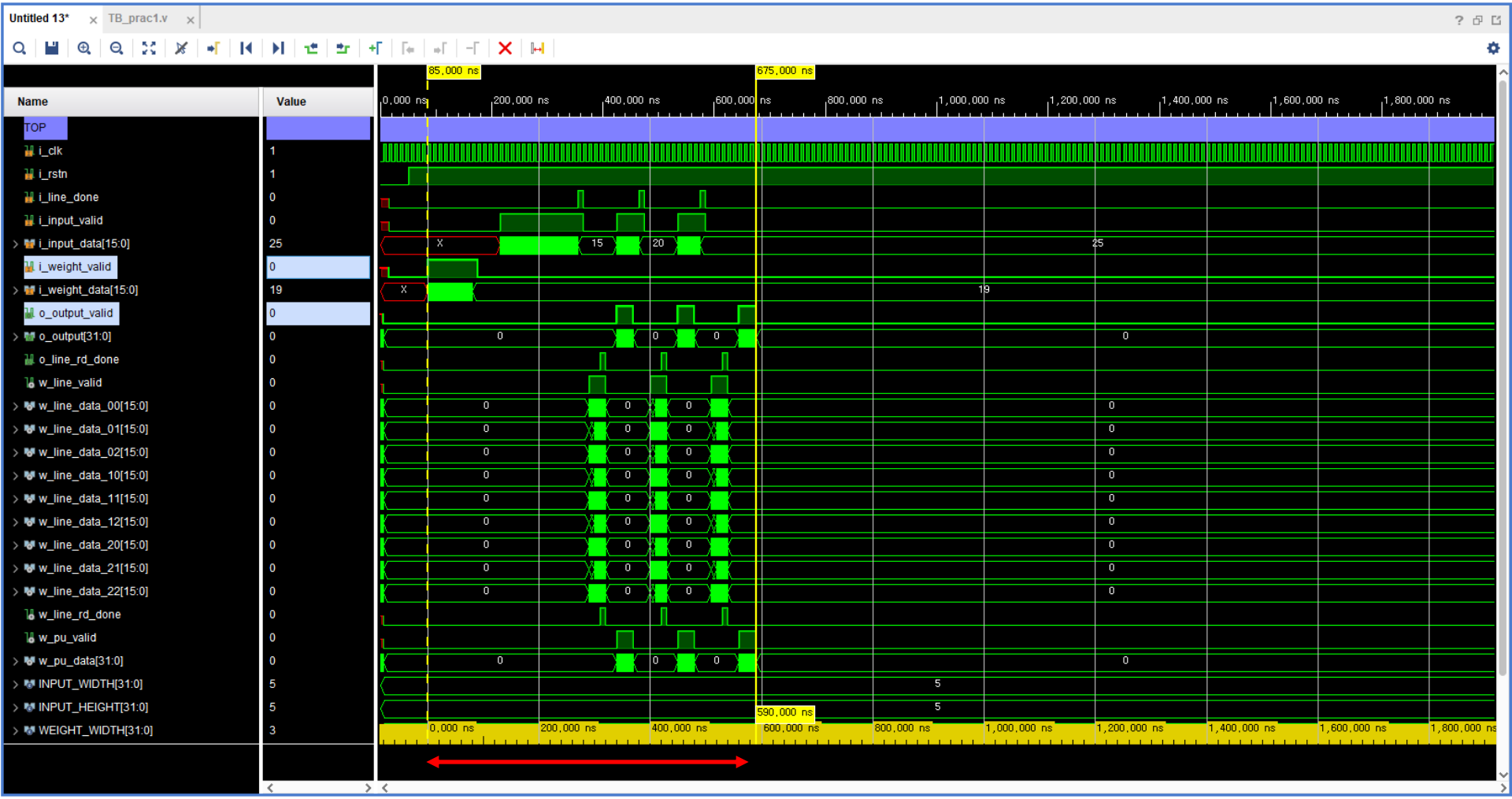
Row 0	1041	1176	1311
Row 1	1716	1851	1986
Row 2	2391	2526	2661





Prob 1. WS with Line Buffer (8)

Waveform (6)



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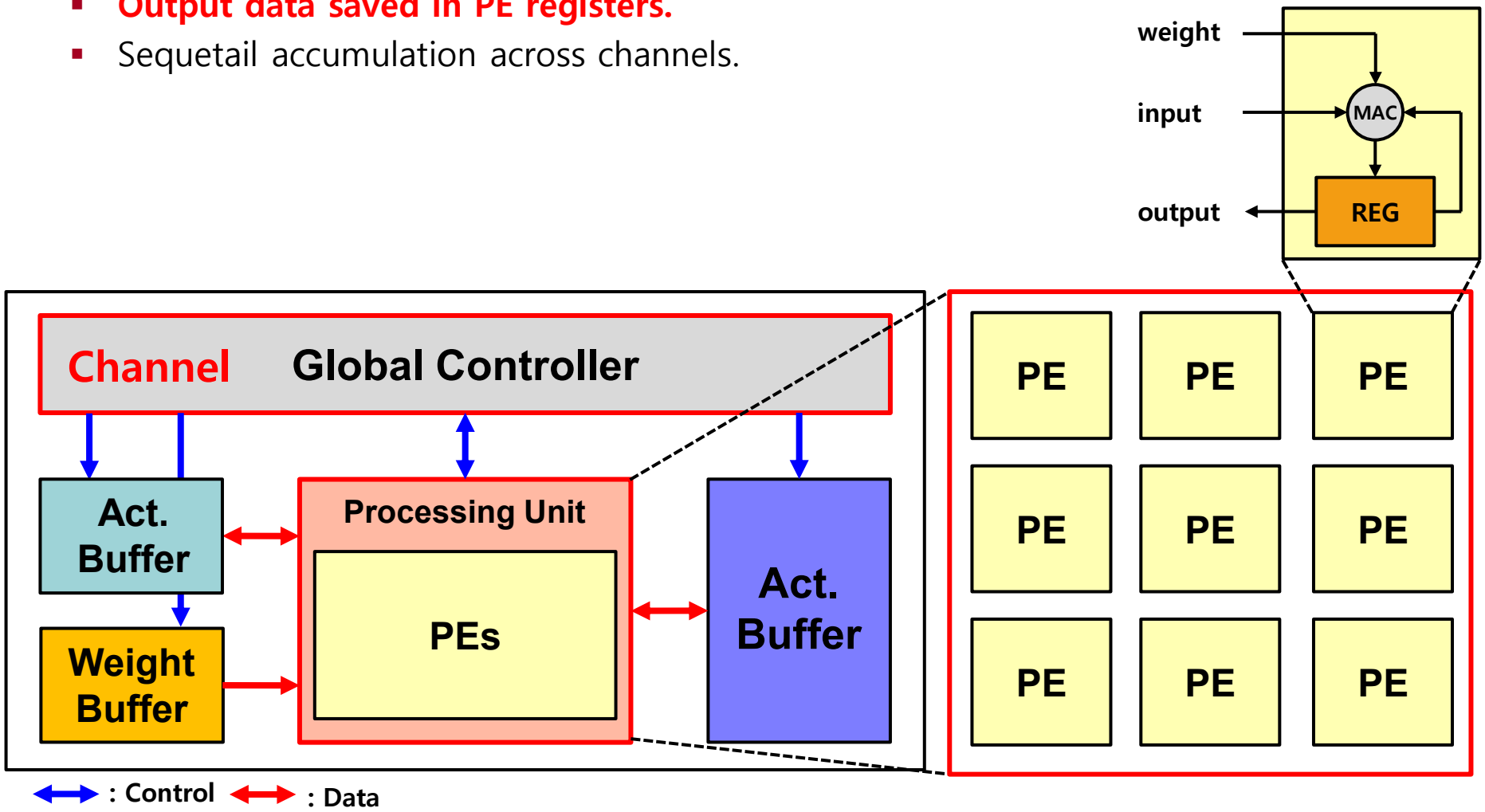
Prob 2. Multi Channel OS (1)

Multi Channel Conv2d Output Stationary.

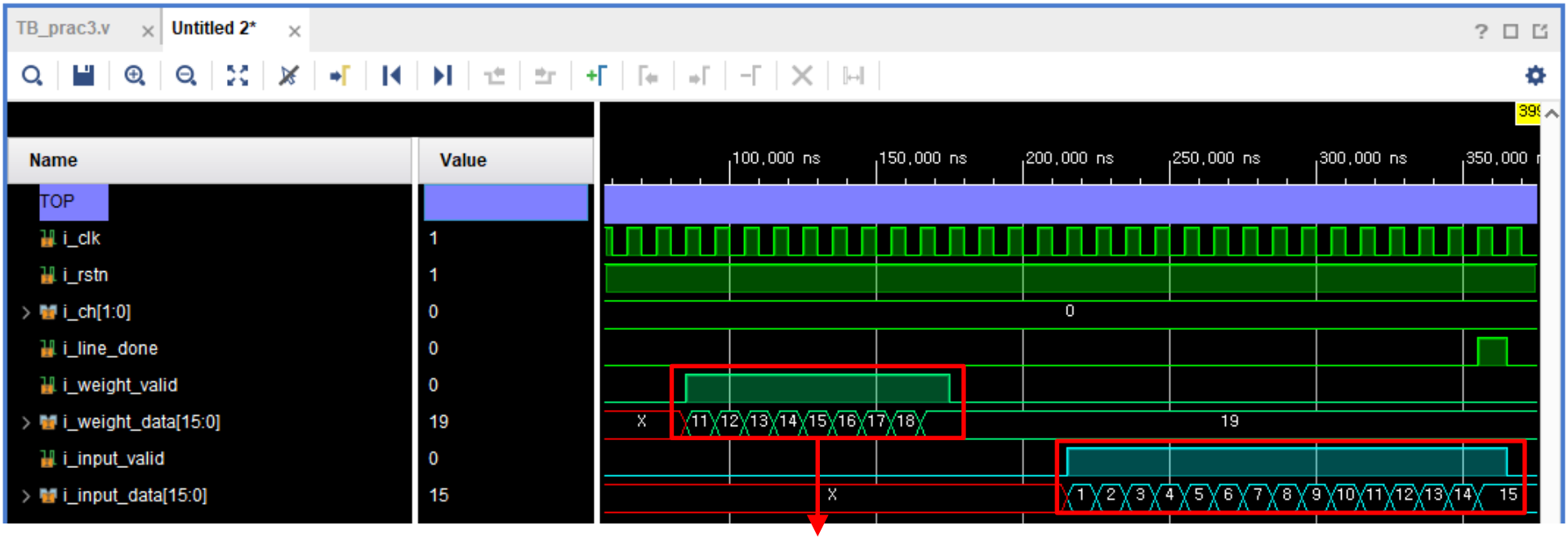
- **Output data saved in PE registers.**
- Sequetail accumulation across channels.



Prob 2. Multi Channel OS (2)

Waveform (1)

- Load weight and input data for **CH1**.

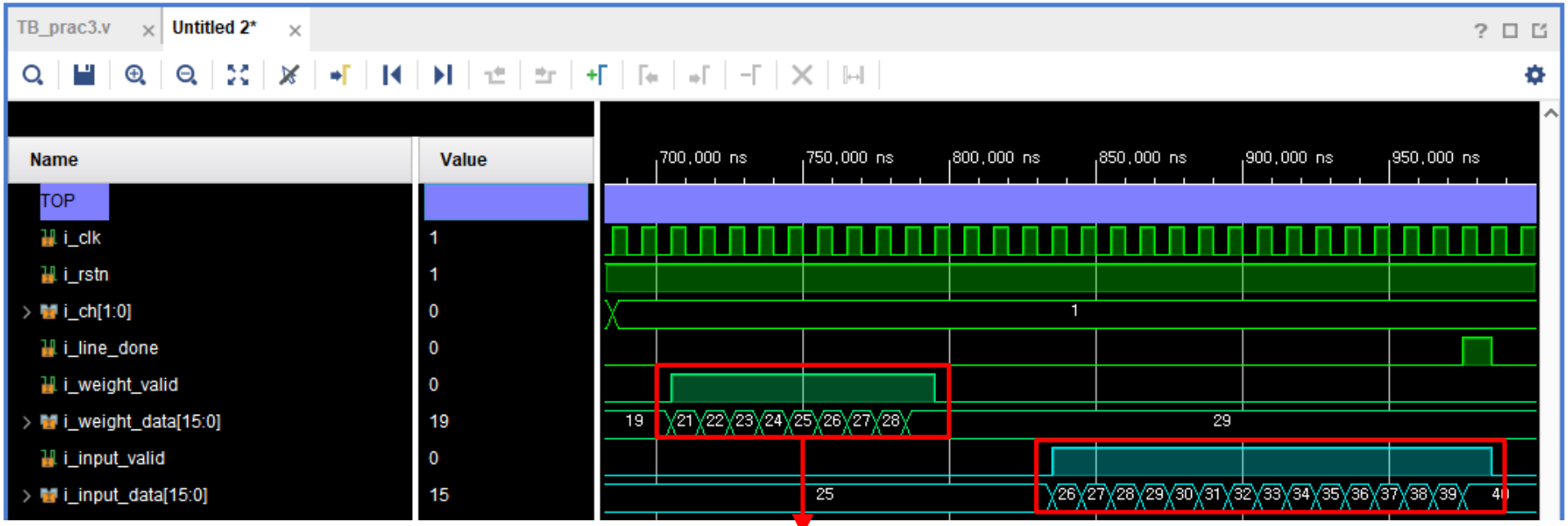


CH1 Weight **CH1 Input**

Prob 2. Multi Channel OS (3)

Waveform (2)

- Load weight and input data for **CH2**.

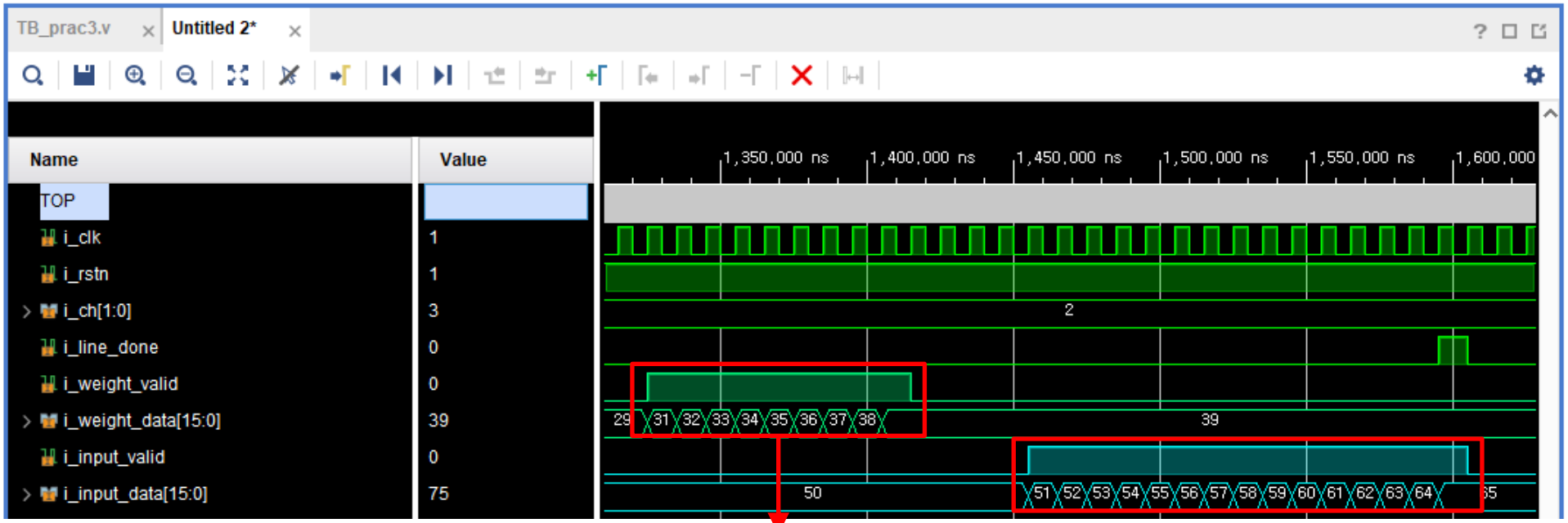


CH2 Weight **CH2 Input**

Prob 2. Multi Channel OS (4)

Waveform (3)

- Load weight and input data for **CH3**.

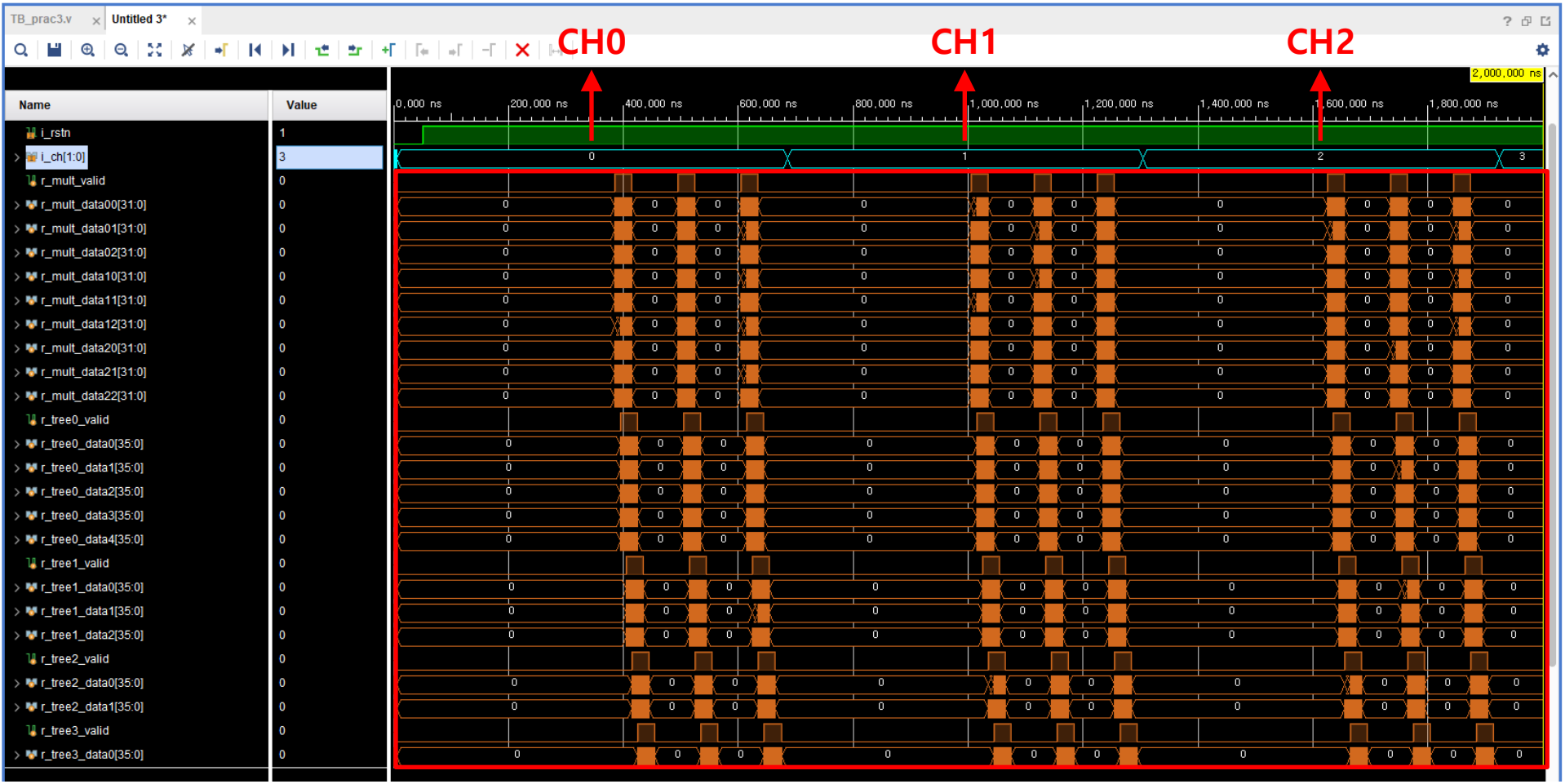


CH3 Weight **CH3 Input**

Prob 2. Multi Channel OS (5)

Waveform (4)

- Output data is stored in registers for partial sum.



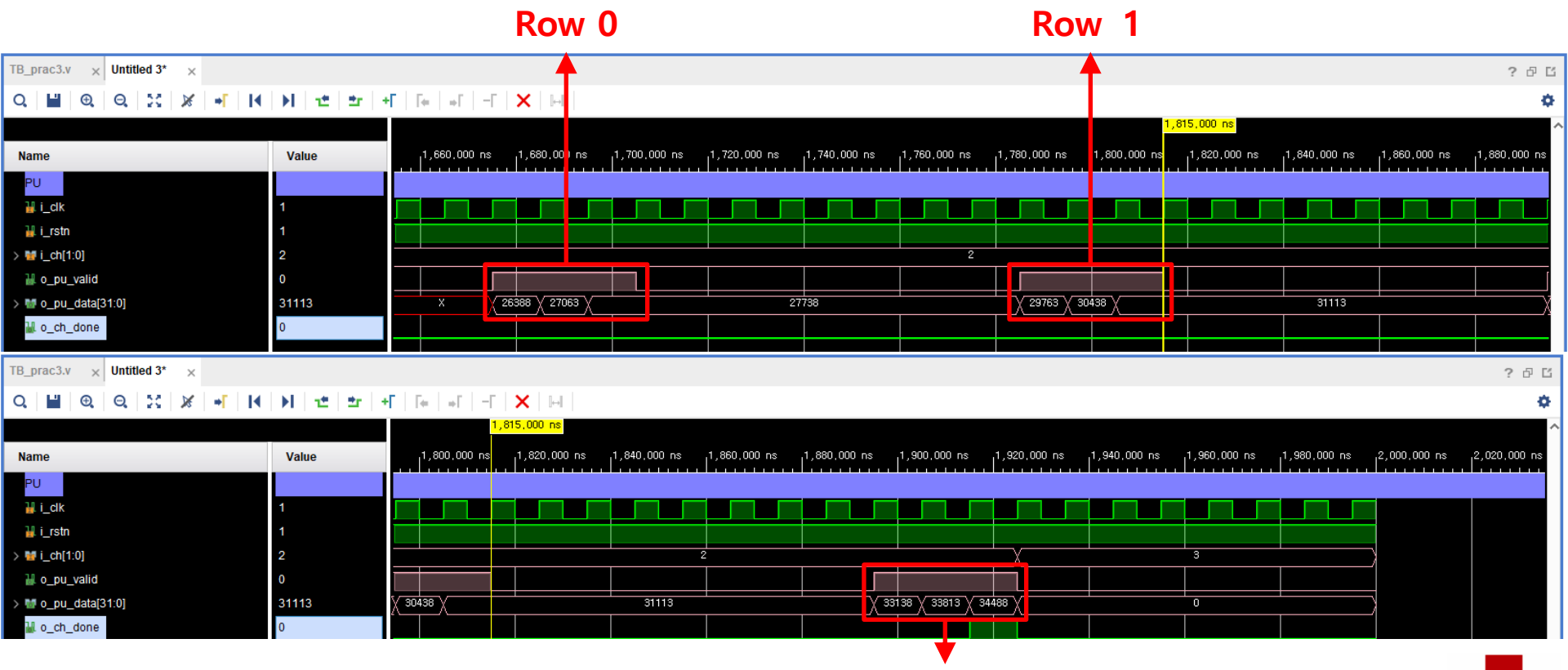
MAC

Prob 2. Multi Channel OS (6)

Waveform (5)

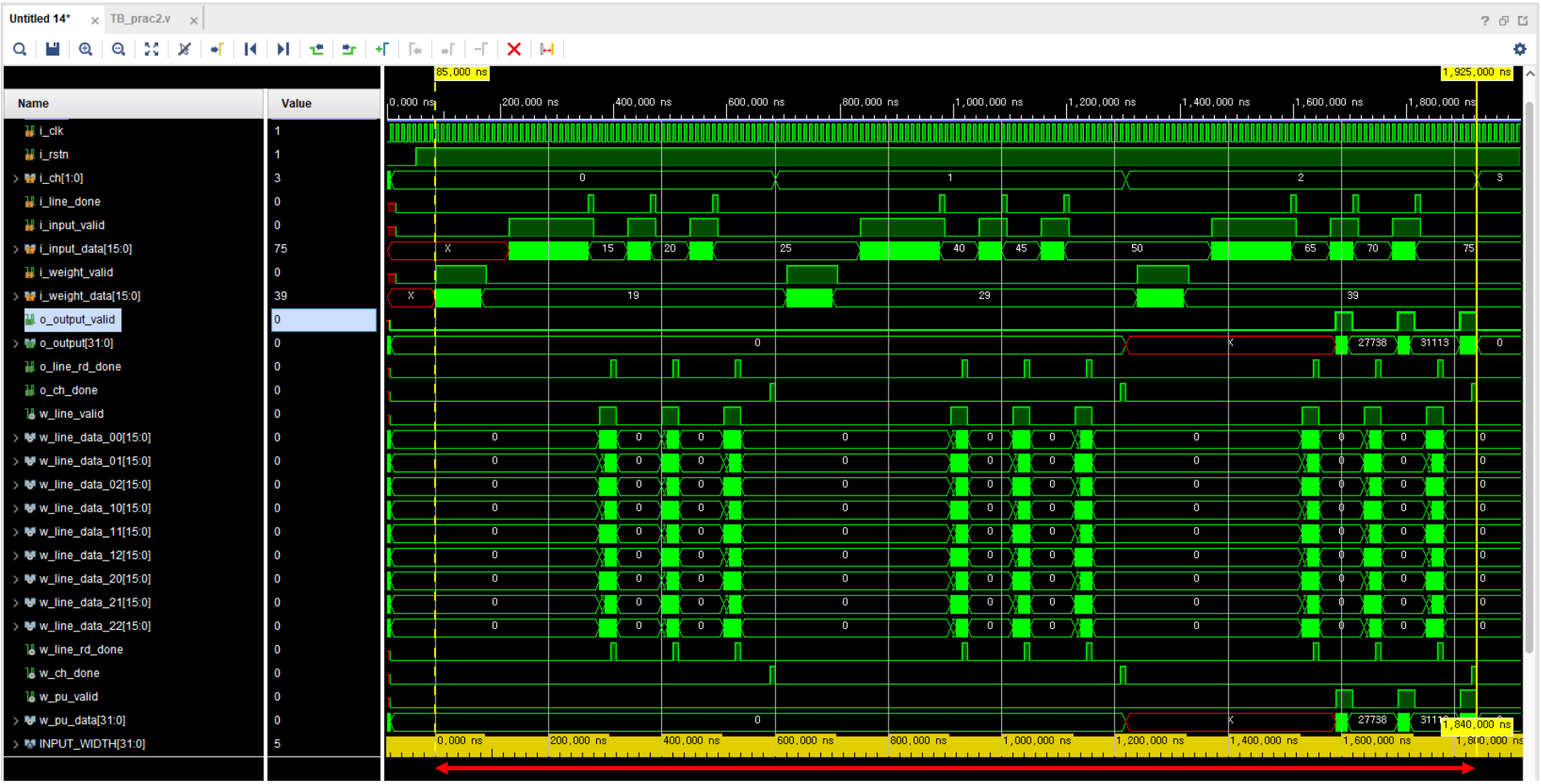
- Output data is stored in registers for partial sum.

Row 0	26388	27063	27738
Row 1	29763	30438	31113
Row 2	33138	33813	34488



Prob 2. Multi Channel OS (7)

Waveform (6)



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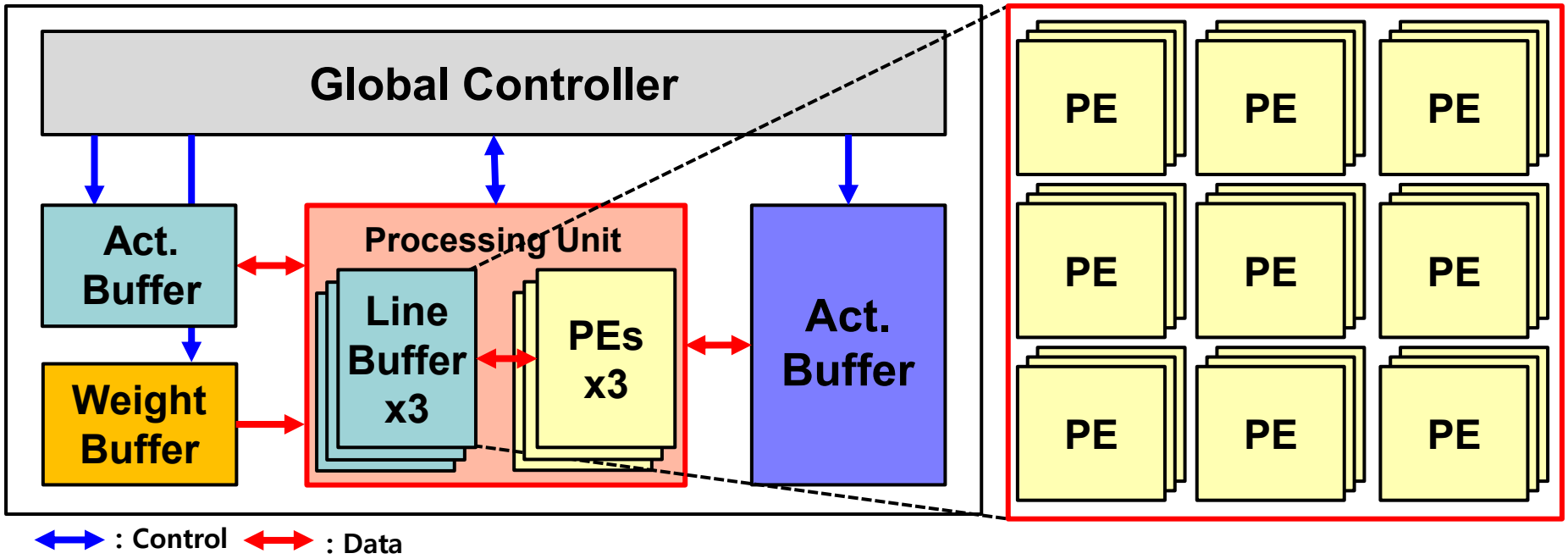
Prob 3. Channel-wise Parallelism (1)

Channel wise parallelism

- **Input data loaded in parallel across channels.**
- Increased line buffers and processing elements.
- Maintains the same processing time as single channel operation.



< Target Hardware Architecture >

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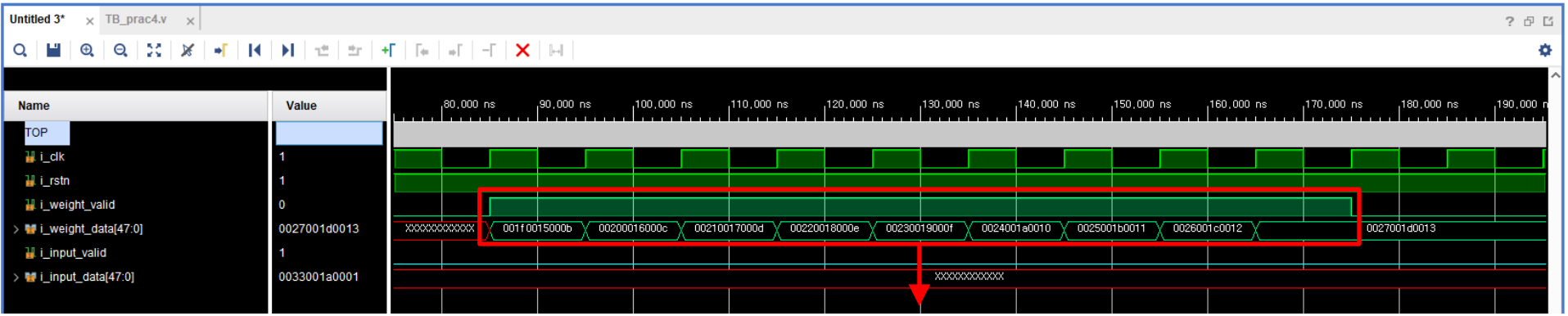
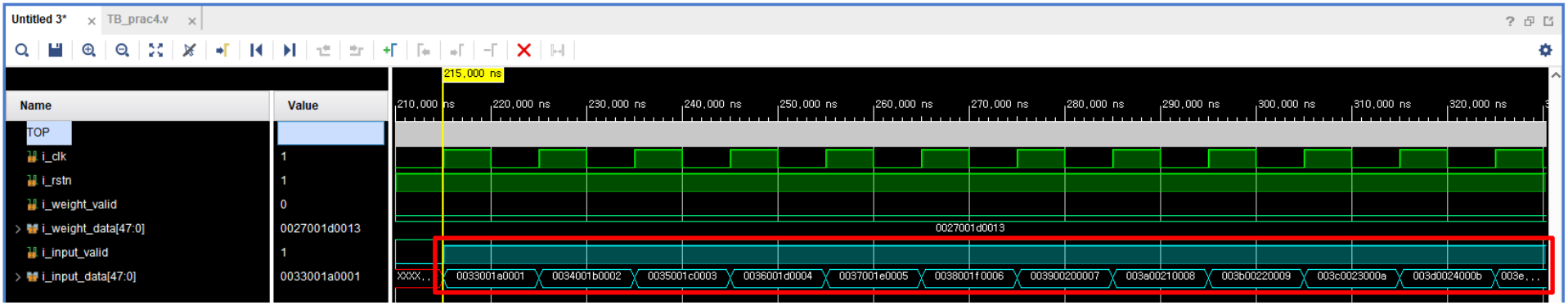
Prob 3. Channel-wise Parallelism (2)

Waveform (1)

- **Load weight and input data across channels.**
- Weight and input data ard loaded in 48-bit units per cycle.



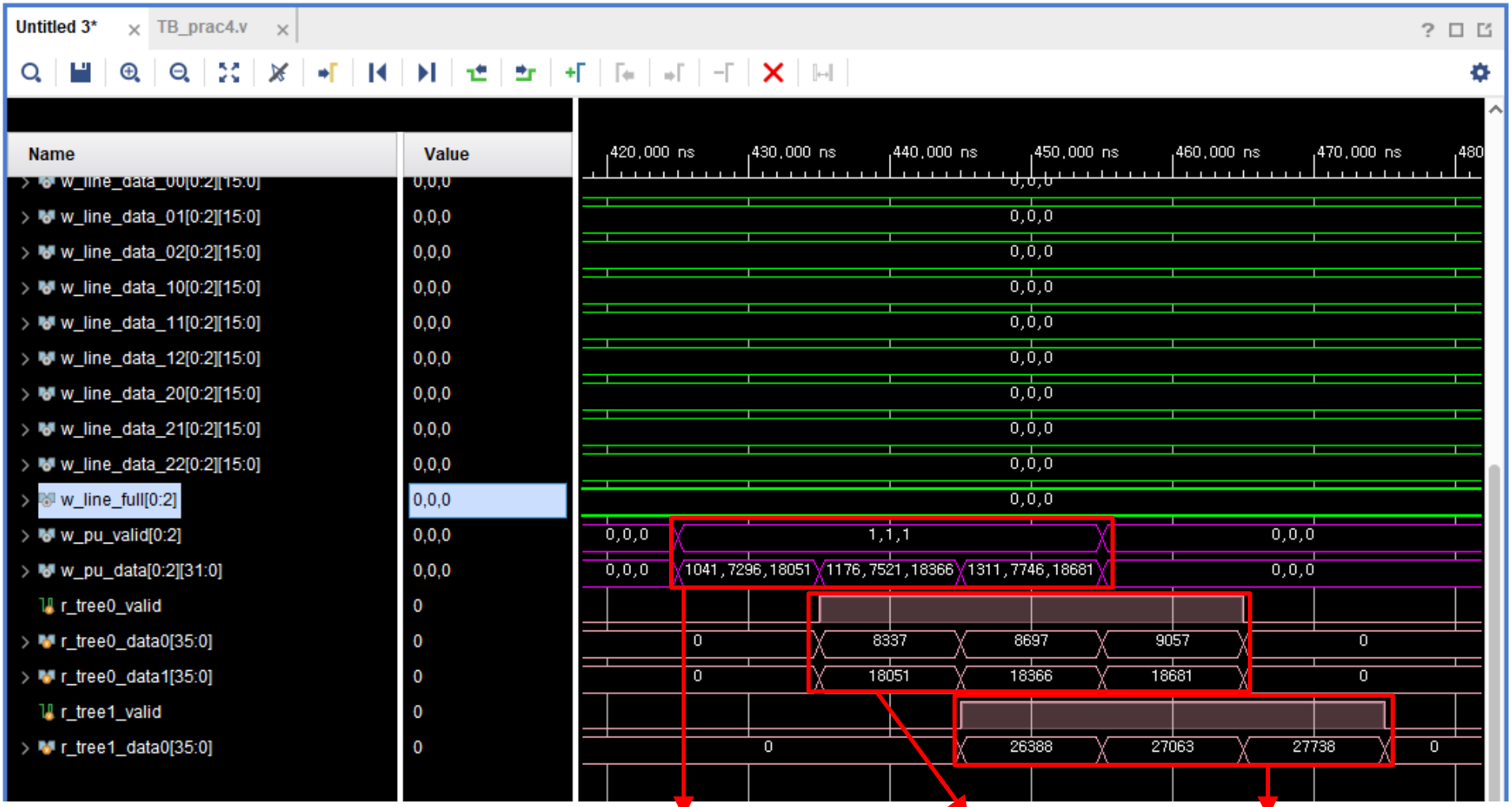
Weight Load



Input Load

Problem 3. CW Parallelism (3)

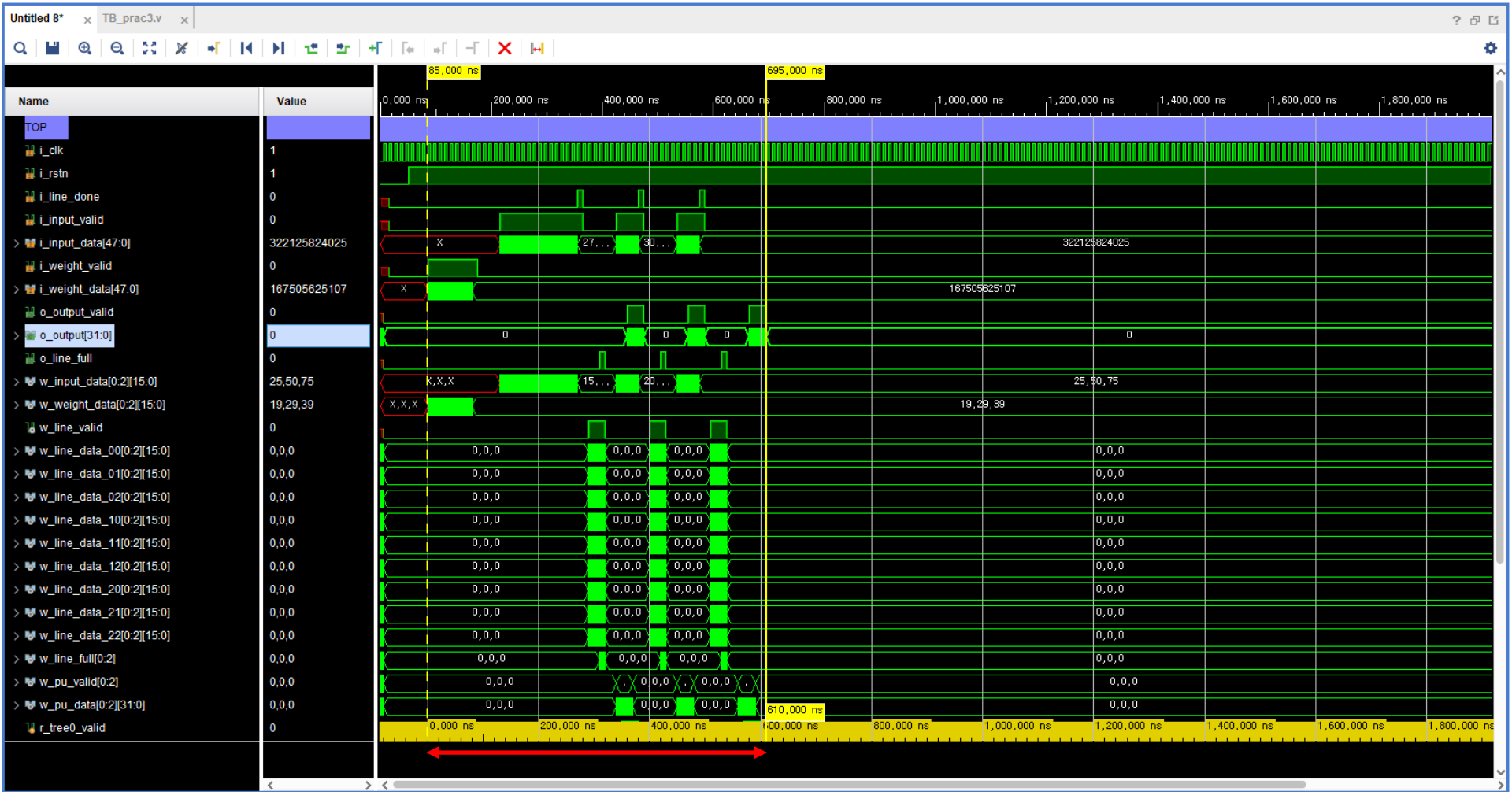
Waveform (2)



Output per Channel ADDER TREE Final Output

Problem 3. CW Parallelism (4)

Waveform (3)



61 Clocks

Thank You!